

# tnbc-92 · results summary (v2)

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## tnbc-92 · results summary (v2 - v3 gpath2vec retrain)

manuscript-current results on the v3 gpath2vec retrain grid (`runs/tnbc-92_v3/`). data version: v3 gpath2vec build  
`fisher_madmean_low_dim512_e5_s1234` (sha 13985cbd...), v3 niche-join  
208,786 niches, test set 35,594 niches / 14 held-out patients / 38  
subarrays.

**v1 baseline** for head-to-head comparison: [`../v1/tnbc92\_results\_summary.md`](#).

**status.** numbers verified from `runs/tnbc-92_v3/eval/*` and per-run  
`metrics_h1_raw.json`. prose interpretation preserved from the v1  
framework (R4 wins H1+H3, late-fusion family wins H2 mc, classical  
baselines lose 30+% on the resolution shift) - all v1 directional claims  
hold on v3. v3 adds **R5\_v3** (AnInfoNCE + late fusion, new method; per-  
dim learnable temperature, one-variable-change vs R1) to the grid.

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## method definitions

R1-R4, R6 and B1-B4 unchanged from v1; see [../v1/tnbc92\\_results\\_summary.md](#) for the full method table. v3 architecture and method-grid axes unchanged; only the ST-side gpath2vec\_niche feature was swapped from v1 to v3 build. **v3 adds one new method:**

| run   | loss                                                                                 | fusion    | ST input                | what it tests                                                                                                                             |
|-------|--------------------------------------------------------------------------------------|-----------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| R5_v3 | AnInfoNCE<br>(per-dim<br>learnable<br>temperature,<br>init =<br>standard<br>InfoNCE) | late MLPs | novae +<br>gpath2vec_v3 | does<br>anisotropic<br>per-dim<br>temperature<br>improve<br>late-fusion<br>alignment<br>over R1?<br>one-<br>variable-<br>change vs<br>R1. |

## H1 - Cross-Modal Alignment (v3 grid)

source: runs/tnbc-92\_v3/eval/H1/summary.json, per-run runs/tnbc-92\_v3/{run}/metrics\_h1\_raw.json.

| run   | method                         | R@1     | R@5     | median rank | AUC   | CKA after | alignment gap |
|-------|--------------------------------|---------|---------|-------------|-------|-----------|---------------|
| R4_v3 | InfoNCE + cross-attn           | 0.00014 | 0.00155 | 3282        | 0.859 | 0.631     | 0.233         |
| R1_v3 | InfoNCE + late                 | 0.00042 | 0.00185 | 6234        | 0.761 | 0.312     | 0.194         |
| R5_v3 | AnInfoNCE + late (new)         | 0.00039 | 0.00177 | 6268        | 0.761 | 0.314     | 0.216         |
| R6_v3 | InfoNCE + late (novae-only ST) | 0.00014 | 0.00079 | 7016        | 0.737 | 0.239     | 0.154         |

| run   | method           | R@1     | R@5     | median rank | AUC   | CKA after | alignment gap |
|-------|------------------|---------|---------|-------------|-------|-----------|---------------|
| R2_v3 | SupCon + late    | 0.00022 | 0.00107 | 7730        | 0.725 | 0.101     | 0.036         |
| B2_v3 | Procrustes       | 0.00008 | 0.00079 | 8070        | 0.706 | 0.126     | 0.161         |
| R3_v3 | Barlow + late    | 0.00000 | 0.00037 | 7644        | 0.674 | 0.306     | 0.143         |
| B1_v3 | CCA              | 0.00006 | 0.00014 | 15554       | 0.544 | 0.076     | 0.007         |
| B4_v3 | random-init MLPs | 0.00006 | 0.00014 | 17936       | 0.496 | 0.120     | -0.0004       |
| B3_v3 | Unaligned PCA    | 0.00000 | 0.00000 | 20688       | 0.441 | 0.126     | -0.053        |

random median-rank baseline =  $\lceil n/2 \rceil = 17,797$  (v3 test pool 35,594).

**v3 vs v1 deltas (AUC):** R4 +0.008, R1 +0.020, R6 +0.004, R2 +0.018, R3 +0.028, B2 +0.003, B1 +0.003, B4 +0.005, B3 -0.002. R5 is new (no v1 comparison). all contrastive runs improved or held; classical baselines essentially unchanged.

**R5\_v3  $\equiv$  R1\_v3 on H1** (AUC 0.761 / 0.761, CKA 0.314 / 0.312).

AnInfoNCE's per-dim learnable scaling barely moved from init (mean log-scale -0.015, std 0.042; only 17/512 dims moved >0.1 in log space). the matched-pair InfoNCE objective doesn't strongly incentivize direction-specific weighting; AnInfoNCE effectively collapses to standard InfoNCE on H1.

**verdict (unchanged from v1).** R4 wins by clean margin on both AUC and CKA. all five contrastive late-fusion runs (R1, R2, R3, R5, R6) plus R4 cross-attn outperform all three classical baselines (B1-B3) on AUC and CKA with no overlap between families. B4 random-init at chance (0.496) confirms architecture alone contributes none of the H1 gain - contrastive training does the bulk, cross-attention adds the rest. B3 unaligned PCA sub-chance with negative gap (sanity floor).

## H2 Part A - niche-level biology coherence (mc\_megacluster, audit-correct)

source: runs/tnbc-92\_v3/eval/H2/mc\_coherence.parquet. KMeans(k=14) on z\_he, ARI vs Wang's per-spot 14-class NMF mc\_megacluster label.

**z\_he is the citable view** (non-circular - H&E never sees mc\_megacluster). z\_st / z\_joint partially circular and reported in the parquet for completeness only.

| run   | archetype<br>ARI<br>(z_he) | mc_megacluster<br>ARI (z_he) | delta<br>archetype → mc | reading                                                           |
|-------|----------------------------|------------------------------|-------------------------|-------------------------------------------------------------------|
| R5_v3 | — (TODO)                   | <b>0.246</b>                 | —                       | top of 3-way tie within KMeans seed noise (vs R6 0.244, R1 0.234) |
| R6_v3 | 0.236                      | <b>0.244</b>                 | +0.008                  | niche-level biology winner (v1)                                   |
| R1_v3 | 0.244                      | <b>0.234</b>                 | -0.010                  | runner-up                                                         |
| B1_v3 | 0.259                      | 0.211                        | -0.048                  | classical, loses leakage advantage                                |
| B2_v3 | 0.306                      | 0.194                        | <b>-0.112</b>           | classical, big leakage drop                                       |
| B3_v3 | 0.307                      | 0.194                        | <b>-0.112</b>           | rotation-equivalent to B2                                         |
| R3_v3 | 0.130                      | 0.159                        | +0.029                  | low                                                               |
| R2_v3 | 0.187                      | 0.152                        | -0.035                  | low                                                               |
| R4_v3 | 0.065                      | <b>0.101</b>                 | +0.036                  | lowest on KMeans (manifold-geometry bias, same v1 finding)        |

| run    | archetype<br>ARI<br>(z_he) | mc_megacluster<br>ARI (z_he) | delta<br>archetype → mc | reading                                                                           |
|--------|----------------------------|------------------------------|-------------------------|-----------------------------------------------------------------------------------|
| raw_he | 0.295                      | 0.164                        | -0.131                  | raw H&E<br>archetype<br>“win” was<br>patient<br>identity                          |
| raw_st | 0.017                      | 0.046                        | +0.029                  | Novae<br>alone<br>barely<br>clusters<br>mc -<br>alignment<br>is doing<br>the work |

**construct-validity receipts on v3:** - classical baselines lose 11+ ARI points archetype → mc (B2/B3 -0.112, B1 -0.048, raw\_he -0.131). quantifies the patient-leakage the audit caught. - **R5 / R6 / R1 in a 3-way tie at the top** within typical KMeans seed-variance (ARI gaps <0.012). late-fusion contrastive (with and without AnInfoNCE, with and without gpath2vec ST anchor) is the right family for niche-level region clustering on tissue. - R4 lowest on KMeans (0.101): same v1 finding - cross-attention compresses to a tight manifold and KMeans assumes globular clusters. metric-induced, not biology absence. **v1 rescue (patient-stratified linear probe at parity) not yet rerun on v3** - flag if needed for manuscript. - alignment provides 5× **lift** over raw ST (0.046 → 0.246 with R5) on niche-level biology coherence.

**archetype-ARI (proposal-literal) reported as diagnostic only**, NOT a biology claim - per construct-validity audit (projects/tnbc-92/evaluation\_question\_audit.md finding #1, NMI(archetype, patient\_id) ≈ 0.89). see also v3 notebook §5 (construct-validity panel), §8b, §8c.

## H2 Part B - compartment cosine contrasts

source: runs/tnbc-92\_v3/eval/H2/compartment\_cosine.parquet  
(regenerated for v3; named-contrast Welch z-test analysis preserved from v1).

| contrast                    | v1 result                           | v3 status |
|-----------------------------|-------------------------------------|-----------|
| TLS vs hi-TIL stroma        | passed R1/R2/R6<br>(z=+3.2 to +5.3) |           |
| TLS vs lo-TIL stroma        | passed R1/R2/R6                     |           |
| Tumor vs hi-TIL / lo-TIL    | inverted in all R-runs              |           |
| Necrosis vs hi-TIL / lo-TIL | inverted in all R-runs              |           |

**v1 interpretation (likely preserved on v3): TLS is the one FTU-like compartment where the proposal's directional prediction held.** annotation-granularity confound on Tumor/Necrosis (large heterogeneous classes dilute matched-pair cosine). see v3 notebook §8c for the FTU framing.

**coverage:** 13/38 v3 held-out test subarrays carry compartment annotations (~35% of test niches, same scope as v1).

## H3 - Pathway Interpretability (gpath2vec arm, proposal-literal)

**this is the proposal-literal H3 test** - per-pathway univariate CCA between each run's shared-latent canonical direction and the **v3 gpath2vec** pathway embeddings (niche cosine profile to embedded Reactome descendants of each named parent). source: `runs/tnbc-92_v3/eval/H3/pathway_cca_gpath2vec_v3/per_pathway_cca.parquet`. embeddings sha256 782fe64f... locked in provenance.

**z\_he view only** (non-circular - v3 gpath2vec is on the ST input, so `z_st` / `z_mean` are circular with the gpath2vec-derived signal; reported in parquet for completeness, not headline). option A = cross-patient held-out 3-patient sub-split; 500 perms; BH-FDR over testable (`set_size`>=5) `z_he` family.

## option A (cross-patient held-out), z\_he z\_A

| run              | Immune<br>(n=76) | ECM<br>(n=15) | Cell<br>Cycle<br>(n=15) | PCD<br>(n=7) | TGF- $\beta$ (n=2,<br>underpowered) | sig<br>testable<br>(/4) |
|------------------|------------------|---------------|-------------------------|--------------|-------------------------------------|-------------------------|
| R4_v3            | 25.4             | 16.0          | 16.5                    | 11.2         | 5.2                                 | 4/4                     |
| B2_v3 /<br>B3_v3 | 17.1             | 9.2           | 10.4                    | 6.7          | 3.1                                 | 4/4                     |
| B1_v3            | 15.2             | 10.4          | 9.8                     | 7.5          | 3.1                                 | 4/4                     |
| R5_v3<br>(new)   | -1.9             | 3.0           | 2.1                     | 3.0          | 0.9                                 | 2/4                     |
| R1_v3            | 2.7              | 1.5           | 1.5                     | -0.6         | 0.2                                 | 1/4                     |
| R3_v3            | 2.3              | 0.1           | 0.7                     | 0.2          | 1.4                                 | 1/4                     |
| R2_v3            | -1.6             | -2.2          | -1.5                    | -1.2         | -1.1                                | 0/4                     |
| R6_v3            | 0.5              | 1.3           | -1.8                    | 0.6          | 0.3                                 | 0/4                     |

TGF- $\beta$  R-HSA-170834 set\_size=2 - **underpowered, excluded from the BH-FDR family**, reported separately. only 4 pathways enter the FDR family on the gpath2vec arm.

subtree coverage on the v3 build: Immune 35%, ECM 79%, Cell Cycle 12%, PCD 17%, TGF- $\beta$  17%. coverage is **not thresholded** (any cut separating PCD~17% from Cell\_Cycle~11% is arbitrary - the F1 anti-pattern); representativeness reported per-pathway, reader-judged.

## verdict (v3 grid, gpath2vec arm)

**R4 wins cross-patient pathway transfer on the proposal-literal test.** 4/4 testable pathways significant at BH-FDR<0.05, z\_A 11.2-25.4. late-fusion contrastive runs largely fail: **R5 2/4** (best of late-fusion family - ECM/CC/PCD pass, Immune actively suppressed at z\_A -1.9), R1/R3 1/4, R2/R6 0/4. classical baselines competitive (4/4 with lower z\_A 3-17). v3 R4 z\_A is **stronger than v1 R4** on the AUCell arm (v1 R4 z\_A 3.8-5.7; v3 gpath2vec-arm R4 z\_A 11.2-25.4) - the v3 gpath2vec build amplifies the cross-patient pathway signal.

**R5 finding (negative result on rescue hypothesis).** AnInfoNCE per-dim temperature was queued (per project\_session\_handoff) as a candidate fix for R4's pathway-direction aliasing. R5 on late-fusion gained 1 sig pathway vs R1 (1/4  $\rightarrow$  2/4) - a small effect. **AnInfoNCE is not the rescue for R4's aliasing**; if the question is "can anisotropic temperature un-alias R4 while preserving cross-patient transfer?" it

needs to be tested on cross-attention directly (R5b - deferred, ~30 min compute, requires patching CrossAttnFusion to register aniso\_log\_scale).

**specificity / aliasing (z\_he)**

R4’s pathway-direction off-diag |mean| ~0.75 (z\_he, computed from canonical directions). consistent with v1 finding that R4’s aliasing - 5 pathway directions collapse toward one biology axis - is the **source** of its cross-patient generalization. one axis is harder to overfit to patient identity than five distinct directions.

routing rule (unchanged from v1): **R4 for cross-patient magnitude on any one pathway; B2 for per-pathway distinguishability across multiple pathways.**

**H3-extended (biology vs patient permutation z-test)**

**Systems-oriented evaluation (verdict holds, same as v1)**

| field                                        | observation                                                                                                                                                                                                                                                                   |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| does one alignment dominate every objective? | <b>no.</b> R4 wins H1 + H3 cross-patient pathway transfer. R5/R6/ R1 (3-way tie within seed noise) win H2 niche-level biology clustering. classical baselines competitive on ECM and fail H1. B1 amplifies patient identity.                                                  |
| conclusion                                   | different alignment architectures optimize different biological objectives. niche-level cluster geometry, cross-modal retrieval, and cross-patient pathway transfer are not co-monotonic. the framework’s value is in exposing the trade-offs and supporting task-conditional |



| field          | observation                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                | routing (see <a href="#">tnbc92_routing_matrix.md</a> ).                                                                            |
| strong methods | R4 (retrieval, cross-patient pathway transfer); R5 / R1 / R6 (niche-level region clustering on tissue)                              |
| baselines      | all baselines lose at least one hypothesis family decisively. B4 random-init at chance confirms architecture alone is insufficient. |

## consolidated proposal-alignment matrix (v3)

| hypothesis               | evaluation question                                                 | v3 observed result                                                                    | strong methods (family) | best method  | why best                                                                                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1 Cross-Modal Alignment | can H&E retrieve matched ST niches better than unaligned/classical? | R4 AUC 0.859, CKA 0.631. R1 / R5 0.761 (tied). R6 0.737. B4 untrained 0.496 (chance). | R4, R1, R5, R6          | <b>R4_v3</b> | cross-modal alignment better than H&E<br>7 H&E tokens<br>mean ST niches<br>ST query selection<br>level morphology<br>product tight effect<br>manipulation the h<br>cross-coupling clear<br>AUC over control<br>family |

| hypothesis                                        | evaluation question                                                   | v3 observed result                                                                                                                                                            | strong methods (family) | best method                                                    | why best                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H2 Niche-level biology coherence (mc_megacluster) | does the aligned space cluster niches by transcriptomic tissue state? | R5 ARI 0.246, R6 0.244, R1 0.234 (3-way tie within KMeans seed noise); R4 0.101 (manifold-geometry bias); classical drop 11+ ARI archetype → mc - quantifies patient-leakage. | R5, R1, R6              | <b>R6_v3</b> (most parsimonious) <b>or R5_v3</b> (nominal top) | mem<br>rand<br>chan<br>conf<br>train<br>arch<br>does<br><br>late-<br>cont<br>pres<br>wide<br>glob<br>man<br>KMe<br>rewa<br>wins<br>pars<br>drop<br>gpat<br>from<br>inpu<br>(Nov<br>spati<br>neigh<br>grap<br>clust<br>state<br>has f<br>load-<br>comp<br>for n<br>biolo<br>per-c<br>temp<br>a ma<br>+0.00<br>with<br>noise |
|                                                   | distinct compartments                                                 | TODO - v3 recompute                                                                                                                                                           | R1, R6 (on TLS)         | <b>R1_v3</b> (v1 result; v3                                    | late-<br>pres                                                                                                                                                                                                                                                                                                              |

| hypothesis                                                    | evaluation question                                                                           | v3 observed result                                                                                                                                                             | strong methods (family)                  | best method        | why best                                                                                                                                                                                                                                      |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H2 Compartment contrasts (FTU coherence)                      | > ambiguous stroma on matched-pair cosine?                                                    | pending; v1 result: only TLS-vs-TIL passes (R1/R2/R6, z = +3.2 to +5.3).                                                                                                       |                                          | recompute pending) | distinction<br>comp<br>cosine<br>sepa<br>had t<br>high<br>(TLS<br>z=+5<br>the c<br>cura<br>anno<br>the c<br>the c<br>comp<br>when<br>prop<br>direc<br>pred<br>held<br>Necr<br>anno<br>are t<br>heter<br>to<br>diffe<br>from<br>stron<br>clear |
| H3 Pathway interpretability (gpath2vec arm, proposal-literal) | do the 5 Reactome targets correlate with shared-latent directions AND transfer cross-patient? | <b>R4 4/4 testable pathways sig, z_A 11.2-25.4</b> (Immune/ECM/CC/PCD). R5 2/4 (best of late-fusion family). R1/R3 1/4; R2/R6 0/4; classical B1/B2/B3 4/4 with lower z_A 3-17. | R4 (overall); B1/B2 (classical fallback) | <b>R4_v3</b>       | cross<br>atten<br>tight<br>colla<br>nam<br>path<br>direc<br>onto<br>robu<br>axis<br>(spec<br>off-d                                                                                                                                            |

| hypothesis       | evaluation question                         | v3 observed result                                                                   | strong methods (family) | best method                                    | why best                                                                                                                                                                                                                                                                                                          |
|------------------|---------------------------------------------|--------------------------------------------------------------------------------------|-------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                             | TGF- $\beta$ underpowered.                                                           |                         |                                                | nich<br>clust<br>geom<br>cross<br>retri<br>cross<br>path<br>trans<br>NOT<br>mon<br>obje<br>R4's<br>alias<br>man<br>trade<br>geom<br>mc) f<br>retri<br>path<br>trans<br>H3).<br>R1's<br>man<br>the r<br>the<br>fram<br>valu<br>expo<br>trade<br>routi<br>right<br>to th<br>meth<br>IS th<br>deliv<br>not a<br>winr |
| Systems-oriented | does one strategy dominate every objective? | no - R4 wins<br>H1 + H3, R5/R6/<br>R1 tie H2 mc,<br>classical lose<br>H1 decisively. | R4, R5,<br>R1, R6       | <b>task-conditional<br/>(no global winner)</b> |                                                                                                                                                                                                                                                                                                                   |

## Methods note

cohort, niche unit (k=6 spatial neighbors, ~1200 cells), evaluation split (subarray-level patient-stratified 85/15, seed=42), ST input composition (novae\_niche + gpath2vec\_niche for R1-R5/B1-B3; novae-only for R6), niche aggregation, MLP architecture: all unchanged from v1. see [../v1/tnbc92\\_results\\_summary.md](#) Methods section for the full details.

**v1 → v3 single change:** gpath2vec\_niche (512-d) is the v3 build (fisher\_madmean\_low\_dim512\_e5\_s1234, sha 13985cbd...) instead of the v1 legacy build. niche-join 208,786 niches (vs v1's 286,250) due to v3 dropping 67,131 niches with no significant pathway under MAD/mean gene selection. test set 35,594 niches (vs v1's 45,661).

**v3-only method addition:** R5\_v3 = AnInfoNCE + late fusion. extends InfoNCE with a per-dim learnable log-scale parameter `aniso_log_scale: nn.Parameter(zeros(shared_dim))` registered on LateFusion when `cfg.anisotropic=True`. loss applies `diag(exp(2*log_scale))` to the bilinear inner product - equivalent to a learned diagonal Mahalanobis metric. `log_scale=0` at init → uniform scale → identical to standard InfoNCE at step 0 (clean warm-start). one-variable-change vs R1; same lr / epochs / seed.

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## pointers

| asset                                                 | path                                                                                                                                  |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| v1 baseline (this doc's predecessor)                  | <a href="#">../v1/tnbc92_results_summary.md</a>                                                                                       |
| v3 retrain plan + memory                              | <code>projects/tnbc-92/v3_phase2_plan.md</code> ,<br>[[ <code>project_v3_retrain_result</code> ]]                                     |
| v3 alignment runs                                     | <code>runs/tnbc-92_v3/{R1,R2,R3,R4,R5,R6,B1,B2,B3,B4}_v3/</code>                                                                      |
| v3 H1 eval                                            | <code>runs/tnbc-92_v3/eval/H1/summary.json</code>                                                                                     |
| v3 H2 eval (archetype + compartment + mc_megacluster) | <code>runs/tnbc-92_v3/eval/H2/summary.json</code> +<br><code>mc_coherence.parquet</code> +<br><code>compartment_cosine.parquet</code> |

| <b>asset</b>                                      | <b>path</b>                                                                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| v3 H3 gpath2vec arm                               | runs/tnbc-92_v3/eval/H3/<br>pathway_cca_gpath2vec_v3/<br>per_pathway_cca.parquet +<br>perm_nulls.parquet +<br>provenance.json |
| v3 H3 AUCell (orthogonal<br>robustness check)     | runs/tnbc-92_v3/eval/H3/<br>pathway_cca/                                                                                      |
| v3 summary notebook + HTML                        | notebooks/final/<br>05_summary_umaps_v3.ipynb +<br>05_summary_umaps_v3.html                                                   |
| construct-validity audit<br>(Cameron's questions) | projects/tnbc-92/<br>evaluation_question_audit.md                                                                             |
| MCP / LangGraph agent design                      | docs/mcp_agent_design.md                                                                                                      |